

Proactive Adaptive Traffic Signal Control, Coordinated with Countdown Timers

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Abstract—Urban traffic networks require sophisticated signal control strategies that adapt to real-time conditions while providing driver information through countdown timers. However, integrating countdown displays with adaptive traffic control presents a fundamental timing prediction challenge: adaptive systems typically make decisions 1-4 seconds before phase changes, while meaningful countdown displays require 10+ seconds of advance notice. A special "1.5 cycle" timing constraint is proposed within the model predictive control (MPC) framework to address this integration challenge, guaranteeing future phase sequences within the prediction horizon to enable countdown functionality while maintaining adaptive optimization benefits. The approach employs the Generalized ARZ (GARZ) macroscopic traffic flow model with data-fitted fundamental diagrams to predict network dynamics. A special target function is proposed to achieve uniform traffic distribution and efficient phase utilization. The MPC optimization uses the Nelder-Mead method with constraint handling for phase duration bounds. Validation experiments using SUMO microsimulation with RESCO benchmark scenarios demonstrate that the macroscopic model achieves sufficient prediction accuracy for control purposes. The integrated control system successfully operates countdown timers with adaptive signal optimization, achieving 10-15% improvements in queue length and delay compared to fixed-time control while providing enhanced driver information systems.

Keywords—*transportation network, transport modeling, macroscopic traffic model, control system, traffic signal control, model predictive control, traffic countdown timer*

I. INTRODUCTION

Urban traffic networks face increasing congestion challenges that demand sophisticated signal control strategies to maximize throughput and minimize delays. The optimization of traffic signal timing has become a critical factor in maintaining efficient transportation flow, with adaptive control systems demonstrating superior performance compared to traditional fixed-time approaches for handling dynamic traffic conditions [1,2,3]. Developing effective traffic signal control systems that can adapt to real-time traffic demands while maintaining network-wide coordination represents an important challenge in transportation engineering.

Adaptive traffic signal control represents one approach to address these challenges by adjusting signal timing based on real-time traffic conditions rather than predetermined fixed schedules. Many existing adaptive algorithms rely on

simplified traffic models [4-10], but a promising development direction involves incorporating nonlinear traffic flow dynamics into control systems. In dense traffic conditions, vehicle interactions create complex phenomena including traffic flow instability [11], wave propagation effects [12], and hysteresis behaviors [13], all of which reduce network throughput capacity. Advanced control systems that account for these nonlinear dynamics offer the potential for performance improvements over conventional approaches.

Traffic light countdown timers represent another distinct approach to improving intersection performance through enhanced driver information systems. These displays have demonstrated measurable impacts on traffic flow characteristics, with studies documenting 22% reductions in start-up lost time and 8-24 additional vehicles per hour per lane in favorable conditions [14, 15]. The effectiveness stems primarily from improved driver preparedness and reduced hesitation during signal transitions, resulting in more consistent headway patterns and enhanced intersection throughput [16].

While both adaptive control systems and countdown timers offer individual benefits for traffic optimization, their integration presents significant technical challenges that have limited practical implementations. Many adaptive control systems, such as SCATS and SCOOT, make timing decisions only 1-4 seconds before phase changes, while meaningful countdown displays require 10+ seconds of advance notice. This timing prediction gap has prevented successful real-world deployments and created an inherent incompatibility between countdown functionality and adaptive optimization benefits [17]. This paper proposes a modified control scheme that addresses these integration challenges by guaranteeing future phase sequences within the prediction horizon, enabling the joint operation of countdown timer displays with adaptive traffic signal control while maintaining system optimization effectiveness. This paper further develops the traffic light control algorithm proposed in the articles [18, 19, 20].

II. TRAFFIC CONTROLLERS AND COUNTDOWN TIMERS

Traffic controllers manage signal phases, each corresponding to permitted movement directions at intersections. Controllers operate according to installed programs that define phase sequences and durations within each cycle. When switching programs, controllers complete the current cycle before implementing the new program.

Assuming fixed phase sequences for each intersection, the vector of programs is denoted as $u = (u^1, \dots, u^v)$ where the program for controller v is $u^v = (\tau_1^v, \dots, \tau_k^v)$. Here τ_k^v represents the duration of phase k , K^v is the number of phases for controller v , and each phase duration is constrained within $[\tau_{\min,k}^v, \tau_{\max,k}^v]$. All permitted program vectors u form the convex set U .

Two control schemes exist for countdown timer operation: continuous countdown throughout the entire phase duration, and partial countdown displaying only the final 10 seconds. This paper implements a continuous countdown scheme as a more challenging setup.

The countdown timers use direct communication protocols such as RS-485 for real-time coordination with the traffic signal controller. This interface enables precise timing control and immediate display updates. Traffic controllers manage semaphores organized into signal groups that control specific movement directions. Each phase contains opening, main, and closing intervals, with signal groups having distinct sequences for different intervals.

The joint use of countdown timers and adaptive traffic control introduces a timing coordination issue. Most adaptive control systems finalize phase changes only a few seconds before they occur, whereas countdown displays require much earlier confirmation typically 10 seconds or more to be meaningful for drivers.

The fundamental issue arises because traffic signals for a given direction may remain unchanged across multiple consecutive phases. In the worst-case scenario where a controller has N phases and a signal group shows the same signal for all phases except one, the system must guarantee execution of $N-1$ phases from signal activation. Therefore, accurate countdown display requires knowledge of the durations of the next $N-1$ phases, where N represents the total number of phases in the controller cycle.

To address this challenge, the control scheme uses periodic program updates where the system must always know which phases will execute in the next cycle. Consequently, new programs must be ready "1.5 cycles" before activation. The "1.5 cycle" requirement means that when a controller switches programs, it must first complete the current cycle, then execute one additional complete cycle, and only then start the new program. Consequently, the control cycle duration T_{cde} must accommodate at least two program cycles:

$$T_{cycle} = 2 \max_{v \in V} (T_{v,\max})$$

where $T_{v,\max}$ is the maximum cycle duration for controller v and V represents the set of all controllers in the network.

This timing constraint has significant control implications: new programs must be calculated and loaded "1.5 cycles" before the program switch occurs. This requirement affects the optimization horizon and computational scheduling of the

control system. The detailed implementation of this constraint is discussed in the MPC section.

III. MACROSCOPIC TRAFFIC FLOW MODEL

A macroscopic approach models traffic flow behavior through spatially averaged quantities. The road state is characterized by density $\rho(x, t)$, average velocity $v(x, t)$, and flow intensity $q(x, t)$, where x represents position along the road and t denotes time. Traffic dynamics are governed by a system of partial differential equations.

The transport network is represented as a directed graph with vertices corresponding to road segments and edges representing movement directions at intersections. Each edge carries weight α_{ij} specifying the proportion of vehicles turning from road i to road j . Traffic controllers regulate movement by permitting or restricting flow along specific edges.

This study employs the generalized ARZ model (GARZ) [21] to describe traffic dynamics. The model's key advantage lies in its data-driven parameter calibration capability. The formulation uses traffic density ρ and the individual driver speed preference parameter ω as independent macroscopic variables:

$$\begin{cases} \rho_t + (\rho v)_x = 0 \\ (\rho \omega)_t + (\rho \omega v)_x = 0 \\ v = V(\rho, \omega) \end{cases}$$

The first equation represents conservation of vehicle count, while the second equation describes the evolution of the speed preference distribution. The velocity function $V(\rho, \omega)$ is calibrated from historical data and relates to flow intensity through $Q(\rho, \omega) = \rho V(\rho, \omega)$, known as the fundamental diagram [22, 23]. Figure 1 illustrates the flow and velocity curves for different values of ω .

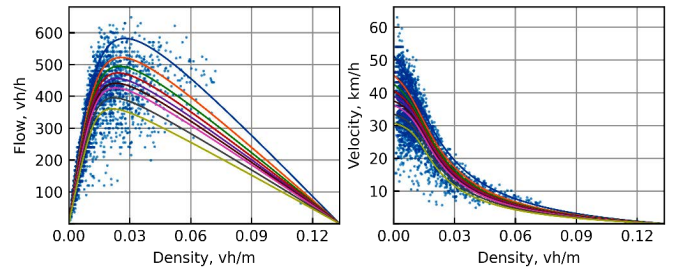


Fig. 1. Data-fitted fundamental diagram for GARZ model.

The fundamental diagram curve is provided by the expression:

$$Q_{a,\lambda,p}(\rho) := \alpha \left(a + (b-a) \frac{\rho}{\rho_{\max}} - \sqrt{1-y^2} \right);$$

$$a = \sqrt{1 + \lambda^2 p^2}; b = \sqrt{1 + \lambda^2 1(-p)^2}; y = \lambda \left(\frac{\rho}{\rho_{\max}} - p \right).$$

Parameter calibration uses historical flow-density observations $(\rho_{data}, Q_{data}) = \{(\rho_i, Q_i) : i = 1, \dots, N_{data}\}$. The parameters α, λ, p are determined by solving:

$$\arg \min_{\alpha, \lambda, p} \left(\beta \left\| Q_{\alpha, \lambda, p}(\rho_{data}) - Q_{data} \right\|_+^2 + (1 - \beta) \left\| Q_{\alpha, \lambda, p}(\rho_{data}) - Q_{data} \right\|_-^2 \right)$$

$$\left\| Q_{\alpha, \lambda, p}(\rho_{data}) - Q_{data} \right\|_+^2 = \sum_{i=1}^{N_{data}} \max\{Q_{\alpha, \lambda, p}(\rho_i) - Q_i, 0\}^2$$

$$\left\| Q_{\alpha, \lambda, p}(\rho_{data}) - Q_{data} \right\|_-^2 = \sum_{i=1}^{N_{data}} \max\{Q_i - Q_{\alpha, \lambda, p}(\rho_i), 0\}^2$$

The parameter β controls the balance between over-prediction and under-prediction penalties. Solving this optimization for various β values yields parameter sets $\alpha_\beta, \lambda_\beta, p_\beta$ and corresponding fundamental diagram functions $Q_\beta(\rho)$.

Since ω represents individual driver speed preference with $\omega = V(0, \omega)$, a mapping from β to ω can be established.

$$V(\rho, \beta) = \begin{cases} Q_\beta(\rho) / \rho, & \rho > 0 \\ Q_\beta'(\rho), & \rho = 0 \end{cases};$$

$$\omega = V(0, \beta) = \frac{\alpha_\beta}{\rho_{\max}} \left(b_\beta - a_\beta + \frac{\lambda_\beta^2 p_\beta}{a_\beta} \right).$$

Now we can apply the mapping $\omega_i = \omega(\beta_i)$, fit polynomial regression on the resulting points and obtain functions $\alpha(\omega), \lambda(\omega), p(\omega)$ for $\omega \in [\omega_{\min}, \omega_{\max}]$. Thus, we obtain required functions $Q(\rho, \omega)$ and $V(\rho, \omega)$ in domain $[0, \rho_{\max}] \times [\omega_{\min}, \omega_{\max}]$. We can extend this domain to $[0, \rho_{\max}] \times R_+$ projecting ω on the segment $[\omega_{\min}, \omega_{\max}]$.

Numerical solution employs Godunov's finite volume method with the computational domain discretized into cells containing averaged traffic parameters. The finite-difference scheme implements:

$$\rho_i^{n+1} = \rho_i^n - \frac{\Delta t}{\Delta x} (Q_{i+1/2}^n - Q_{i-1/2}^n);$$

$$y_i^{n+1} = y_i^n - \frac{\Delta t}{\Delta x} (\omega_{i-1}^n Q_{i+1/2}^n - \omega_i^n Q_{i-1/2}^n);$$

$$\omega_i^n = y_i^n / \rho_i^n.$$

Inter-cell flows $Q_{i\pm 1/2}^n$ are computed by solving Riemann problems using the sending and receiving function formalism [24].

For adjacent cells $U_L = (\rho_L, \omega_L)$ and $U_R = (\rho_R, \omega_R)$, the sending function $S(U_L, U_R)$ determines the maximum flow capacity from cell U_L , while the receiving function $R(U_L, U_R)$ specifies the maximum flow that cell U_R can accept. The inter-cell flow is constrained by $Q = \min\{S, R\}$.

The GARZ model requires an intermediate state $U_M = (\rho_M, \omega_M)$ for proper Riemann problem solution. When $\omega_L < v_R = V(\rho_R, \omega_R)$, the intermediate state is $U_M = (0, \omega_L)$. Otherwise, it satisfies:

$$\rho_M : v_R = V(\rho_M, \omega_L); \quad \omega_M = \omega_L.$$

The sending and receiving functions require the critical density and maximum flow:

$$\rho_c(\omega) = \arg \max_\rho Q(\rho, \omega); \quad Q^{\max}(\omega) = \max_\rho Q(\rho, \omega).$$

The GARZ sending and receiving functions are defined as:

$$S(\rho_L, \omega_L) = \begin{cases} \rho_L \omega_L, & \rho_L \leq \rho_c(\omega_L) \\ Q^{\max}(\omega_L), & \rho_L > \rho_c(\omega_L) \end{cases};$$

$$R(\rho_M, \omega_L) = \begin{cases} Q^{\max}(\omega_L), & \rho_L \leq \rho_c(\omega_L) \\ \rho_M \omega_L, & \rho_L > \rho_c(\omega_L) \end{cases}.$$

Intersection dynamics require modeling traffic flow splitting and merging operations. The directional sending function from road i to road j is defined as $S_{ij} = \alpha_{ij} \delta_{ij} S_i$, where α_{ij} represents the turning proportion, δ_{ij} indicates movement permission, and S_i is the upstream sending capacity. Traffic splitting uses the directional sending function S_{ij} with standard receiving function calculations for the downstream road segment.

Flow merging at road entrances is formulated as a constrained optimization problem:

$$\max_{f_{ij}} \left(\sum_{i=1}^M \sum_{j=1}^N f_{ij} \right); \quad 0 \leq f_{ij} \leq S_{ij}; \quad \sum_{i=1}^M f_{ij} \leq R_j(\rho_{j-}, \omega_{j-}).$$

For M incoming flows, f_{ij} represents the flow from road i to road j . The receiving function R_j requires the entrance state $U_{j-} = (\rho_{j-}, \omega_{j-})$ determined by:

$$\omega_{j-} = \frac{\sum_{i=1}^M \omega_i f_{ij}}{\sum_{i=1}^M f_{ij}}; \quad v_{j-} = \min\{\omega_{j-}, v_j\};$$

$$\rho_{j-} : v_{j-} = V(\rho_{j-}, \omega_{j-}).$$

The model incorporates non-cooperative driver behavior at intersections by assuming that flow allocation is proportional to capacity ratios. The share β_{ij} of each turning movement in the total flow is proportional to the ratio of the directional sending capacity S_{ij} to the total available capacity. The total entrance flow $F_j = \sum_{i=1}^M f_{ij}$ is distributed according to:

$$f_{ij} = \beta_{ij} F_j; \quad \beta_{ij} = \frac{S_{ij}}{\sum_{i=1}^M S_{ij}}.$$

This formulation enables analytical solution of the optimization problem through direct substitution of the flow allocation expressions.

IV. MODEL PREDICTIVE CONTROL

We will control the system at discrete time moments $t_i = i\Delta t$. It is assumed that at the moment t_i we know the current state of the transport network s_i and the input flows d_i . Using the transport network model F , we can predict the behavior of the system within some prediction horizon T for the chosen control u_i . Based on this prediction we can estimate the value of the target function $J_F(s_i, d_i, u_i)$. Then the choice of the optimal control can be formulated as a constrained optimization problem $\arg \min_{u_i \in U} J_F(s_i, d_i, u_i)$.

The integration of countdown timers imposes specific timing constraints on the MPC implementation. The prediction horizon T must be equal to or longer than the control cycle duration T_{cycle} to ensure adequate optimization scope. During the current control cycle starting at time t , the system must calculate the program for the next control cycle starting at $t + T_{cycle}$. To satisfy countdown timer requirements, new programs must be computed and ready "1.5 cycles" before any controller reaches its cycle end, creating an optimization deadline constraint that affects the computational scheduling rather than the optimization problem formulation itself.

In this paper, we use a specially designed target function that estimates the risk of excessive traffic concentration on one part of the transport network. It is assumed that a more uniform load distribution increases the average road capacity and reduces the probability of traffic jam buildup. To implement this strategy, the following target function is used:

$$\begin{aligned} CDS(u) &= \frac{1}{T|L|} \sum_{t=1}^T \sum_{l \in L} \left(\frac{\bar{\rho}_l^t}{\rho_c(\omega_{\max})} \right)^2; \\ WWP(u) &= \frac{1}{T|C|} \sum_{t=1}^T \sum_{c \in C} (1 - \delta_c^t) (\bar{\rho}_{c_{in}}^t - \bar{\rho}_{c_{out}}^t); \\ J(u) &= \alpha CDS(u) + \beta WWP(u) + \gamma \|u_{\min} - u\|_2^2 \end{aligned}$$

For each control cycle, this objective function corresponds to $J(u) = J(s_i, d_i, u)$ where s_i represents the current network state and d_i the input flows at time t_i .

$CDS(u)$ (critical density score) reflects the accumulation of the penalty for excessive density on the road in the transport network. We get an estimate of congestion by comparing the average density $\bar{\rho}_l^t$ on road $l \in L$ at time t with the critical

density ρ_c , and then squaring the resulting ratio. Thus, we will penalize more when the average density is greater than the critical density.

$WWAP(u)$ (Wait Weighted Aggregate Pressure) evaluates the efficiency of phase utilization by the traffic signal controller. The indicator $\delta_c^t = 1$ when direction $c \in C$ is open at time t , and 0 otherwise. For each direction c at the intersection, the average flow density at the entrance $\bar{\rho}_{c_{in}}^t$ and at the exit $\bar{\rho}_{c_{out}}^t$ are evaluated. When the entrance of a direction is nearly full while the exit is free, and the direction is closed, a large penalty is added to discourage inefficient signal timing.

The third term is a standard regularization component, where $u_{\min}$ represents the minimum allowable phase durations that promote baseline signal timing stability.

The constrained optimization problem is solved using the Nelder-Mead method with out-of-bound penalty, where test points that violate the constraint set U are projected back onto the feasible region defined by the phase duration bounds. The GARZ model provides the traffic flow predictions necessary to evaluate the objective function $J(u)$ over the prediction horizon.

V. NUMERICAL EXPERIMENTS

To test the proposed control algorithm, the software package of microscopic traffic simulation SUMO [25] was used. In the framework of the experiment, the real transport network data are substituted by the synthetic data generated by SUMO. The benchmarks set RESCO [26] was used as test scenarios. The 'libsumo' library was used to collect measurements of road detectors (inductive loops) and to control traffic lights controllers in SUMO-simulation. Figure 2 demonstrates the road networks for some test scenarios from the benchmark set.

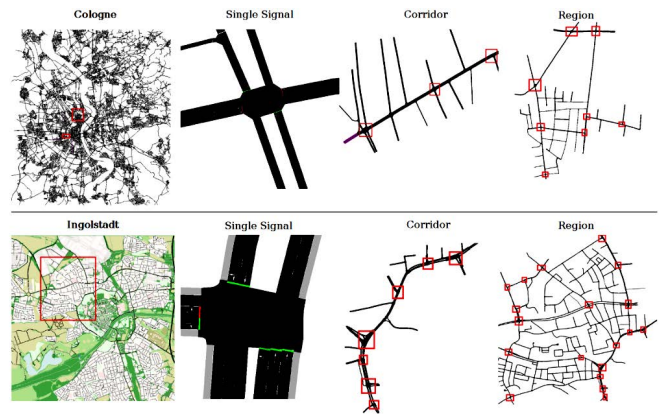


Fig. 2. Test sections of the transport network.

The calibration process for the macroscopic model involved several key steps to ensure accurate representation of traffic dynamics. The SUMO network graph was converted to the format required for macroscopic simulation, preserving the

topological structure while adapting road segment discretization and intersection connectivity.

Roads in the network were grouped by maximum speed limits to establish parameter estimation categories. For each group, fundamental diagram data was collected using SUMO's edgeData-output functionality, which measures flow (vehicles/hour) and density (vehicles/km) at regular intervals. The data collection methodology followed the approach described in the SUMO documentation [27], employing circular route scenarios with evenly spaced vehicles and systematically decreasing departure speeds to capture the complete flow-density relationship across free-flow, congested, and capacity conditions.

The fundamental diagrams were constructed by fitting the GARZ model parameters to the collected flow-density data points using the optimization procedure described in the macroscopic model section. To determine the turning coefficients α_{ij} , baseline simulations were executed using default traffic signal plans, and vehicle distribution patterns at intersections were analyzed to calculate the proportion of vehicles turning from road i to road j .

For model validation, vehicles trajectories provided by SUMO were processed using kernel density estimation (KDE) to reconstruct continuous macroscopic traffic fields for direct comparison with the GARZ model predictions. First, a SUMO simulation was run, and trajectories data was collected. Then, we run the proposed macroscopic simulation and recorded traffic parameters. For each lane, the RMSE error of the model prediction at the detector location was calculated. The estimate of the total error of the transport network model is determined by averaging the error over all nonempty lanes. The percentage of error to the maximum possible value provided by the fundamental diagram is given in brackets. Error estimates for this model are given in Table 1.

The control algorithm was evaluated against default fixed-time plans provided in test scenarios. The experimental parameters were configured with a control cycle duration of 300 seconds and a prediction horizon of 450 seconds, ensuring sufficient look-ahead time for effective optimization while satisfying the countdown timer constraints described earlier. The objective function weights were set to $\alpha = 1$, $\beta = 0.4$, and $\gamma = 0.05$, balancing critical density penalties, wait-weighted pressure considerations, and regularization terms.

TABLE I. MODEL ERROR (RMSE) COMPARED TO SUMO SIMULATION

Scenario	Intensity, vh/h	Density, vh/km	Velocity, km/h
Arterial4x4	169.8 (7.9%)	34.4 (17%)	15.7 (28.3%)
Grid4x4	59.9 (2.8%)	5.8 (0.4%)	3.4 (6.8%)
Cologne1	194.0 (9.0%)	33.9 (16.9%)	15.5 (24.9%)
Cologne3	126.1 (5.8%)	17.4 (8.7%)	5.9 (12.2%)
Cologne8	122.7 (5.7%)	8.5 (4.3%)	2.8 (6.5%)
Ingolstadt1	210.7 (9.8%)	25.2 (12.6%)	9.0 (21.4%)
Ingolstadt7	154.7 (7.2%)	35.8 (17.9%)	12.9 (26.2%)
Ingolstadt21	166.8 (7.7%)	13.7 (6.8%)	5.2 (10.3%)

For the constrained optimization problem, the Nelder-Mead simplex method [28] was implemented with out-of-bound

penalty and bound projection to handle the feasible region defined by minimum and maximum phase duration constraints. The algorithm iteratively refines control solutions by evaluating the objective function at simplex vertices and applying reflection, expansion, contraction, and shrinkage operations to converge toward optimal phase timing. To ensure robust performance assessment, each test scenario was executed 5 times with different random seeds, and the results were averaged to account for stochastic variations in traffic demand and simulation dynamics. The RESCO benchmark scenarios provided standardized testing conditions across different network topologies and traffic loading patterns. Table 2 shows the results of the proposed control algorithm and their comparison with fixed plans.

TABLE II. MODEL PERFORMANCE IN DIFFERENT SCENARIOS

Scenario	Delay	Queue
arterial4x4	652.6/546.4 (16.3%)	16.9/16.0 (5.3%)
grid4x4	89.9/87.5 (2.7%)	0.7/0.7 (5.1%)
cologne1	37.6/36.7 (2.5%)	4.6/4.3 (4.7%)
cologne3	38.3/32.3 (15.8%)	1.6/1.2 (24.5%)
cologne8	42.9/39.3 (8.3%)	0.6/0.5 (18.5%)
ingolstadt1	28.2/21.1 (25.0%)	2.8/1.8 (37.1%)
ingolstadt7	60.3/54.6 (9.4%)	1.4/1.1 (16.9%)
ingolstadt21	128.9/115.9 (10.1%)	0.7/0.6 (15.1%)

CONCLUSION

This paper presents a novel integration of countdown timers with adaptive traffic signal control, addressing the timing prediction challenge that has prevented practical implementations. The key contribution lies in developing a "1.5 cycle" timing constraint that guarantees future phase sequences within the prediction horizon, enabling countdown display functionality while maintaining adaptive optimization benefits. The proposed MPC framework incorporates a specialized objective function to achieve uniform traffic distribution and efficient phase utilization.

The approach demonstrates practical implications for real-world deployment, providing urban traffic management systems with a viable solution that combines adaptive signal optimization with enhanced driver information. Validation experiments confirm that the macroscopic traffic model achieves sufficient prediction accuracy across diverse network scenarios. However, the system faces limitations including growing computational complexity with larger networks, though the Ingolstadt21 scenario demonstrates scalability to substantial urban networks. The approach also depends critically on accurate traffic data for proper GARZ model calibration, which may require significant infrastructure investment for sensor deployment and data collection.

Future research should explore alternative optimization approaches to handle the inherently non-convex target function more efficiently. Additionally, investigating less restrictive timing constraints, such as guaranteeing only N-1 phases instead of the current "1.5 cycle" requirement, could reduce computational overhead while maintaining countdown timer functionality. The modular framework design enables flexible replacement of transportation models, objective functions, and

optimization algorithms, setting multiple directions for continued development.

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